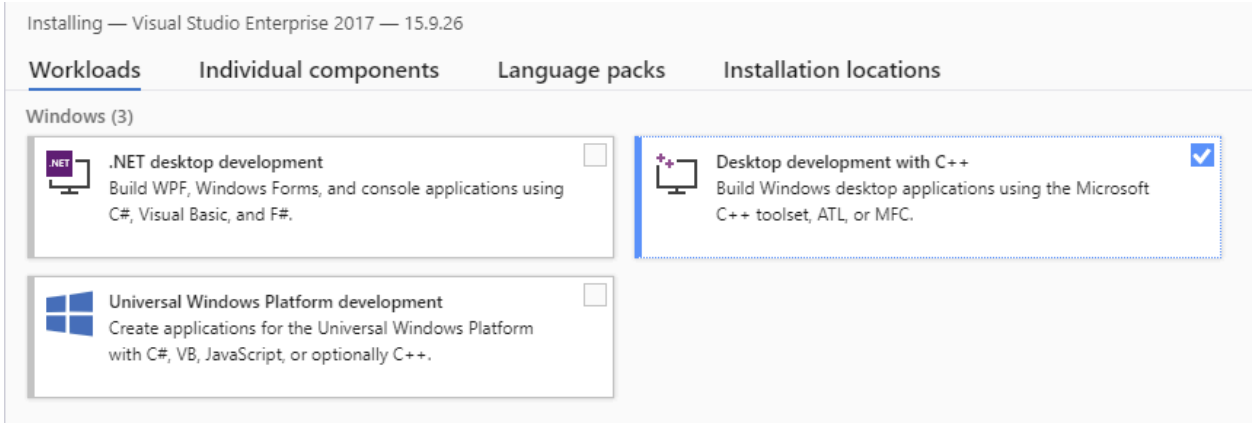
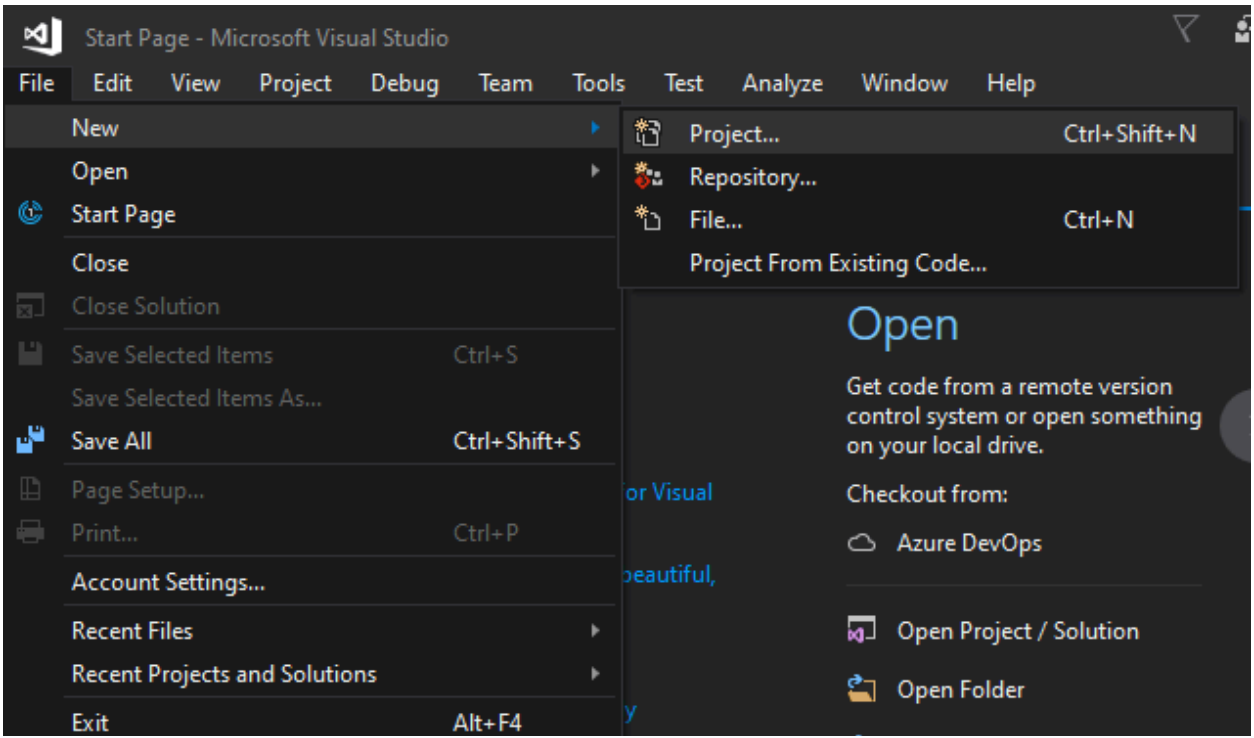
CPS 511 Fall 2020
OpenGL Setup Guide (Visual Studio 2017)

NOTE: you may want to use the latest version of Visual Studio. These instructions will still be useful but there may also be some small differences. You can other documents like this one online.

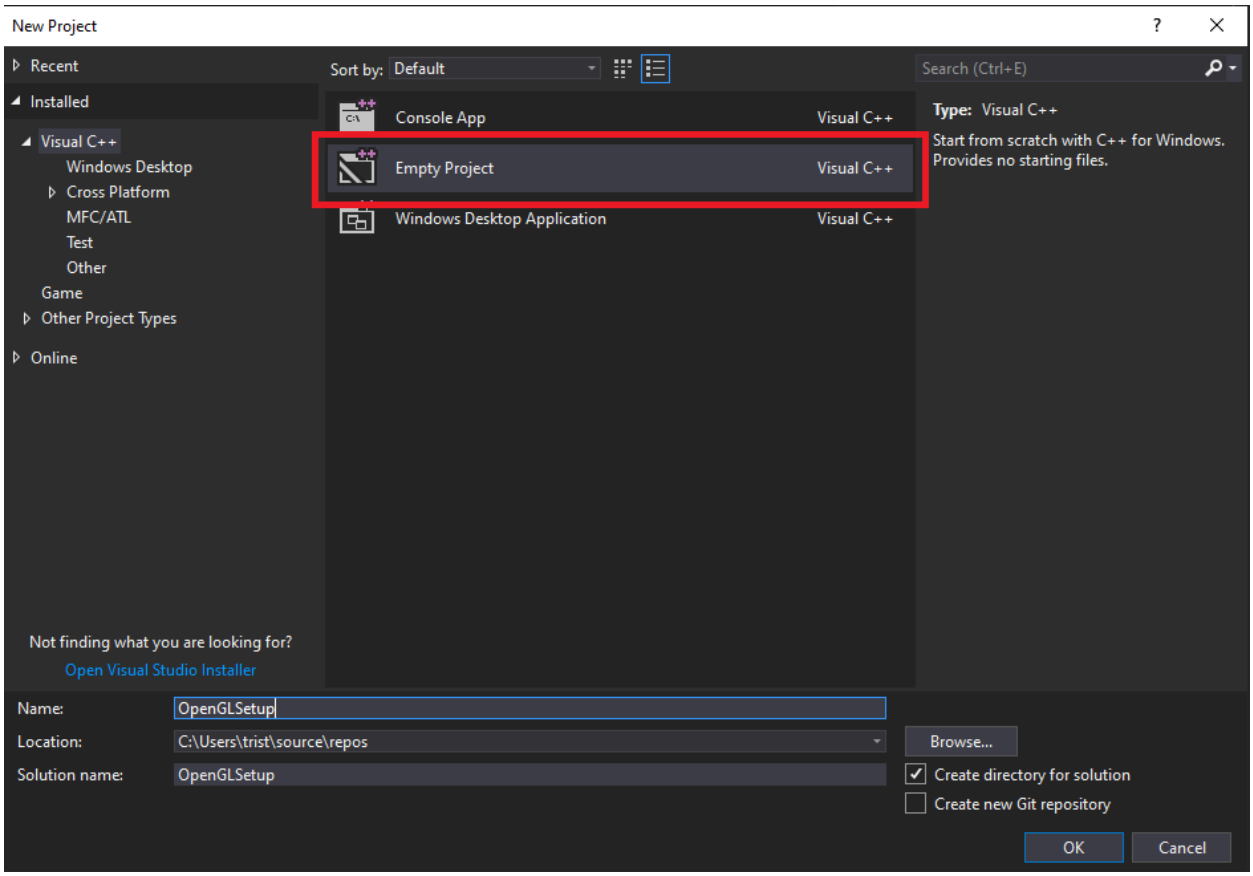
When Installing VS2017 make sure you have the ***Desktop development with C++*** workload checked. This will install the Windows SDK which comes with the OpenGL library.



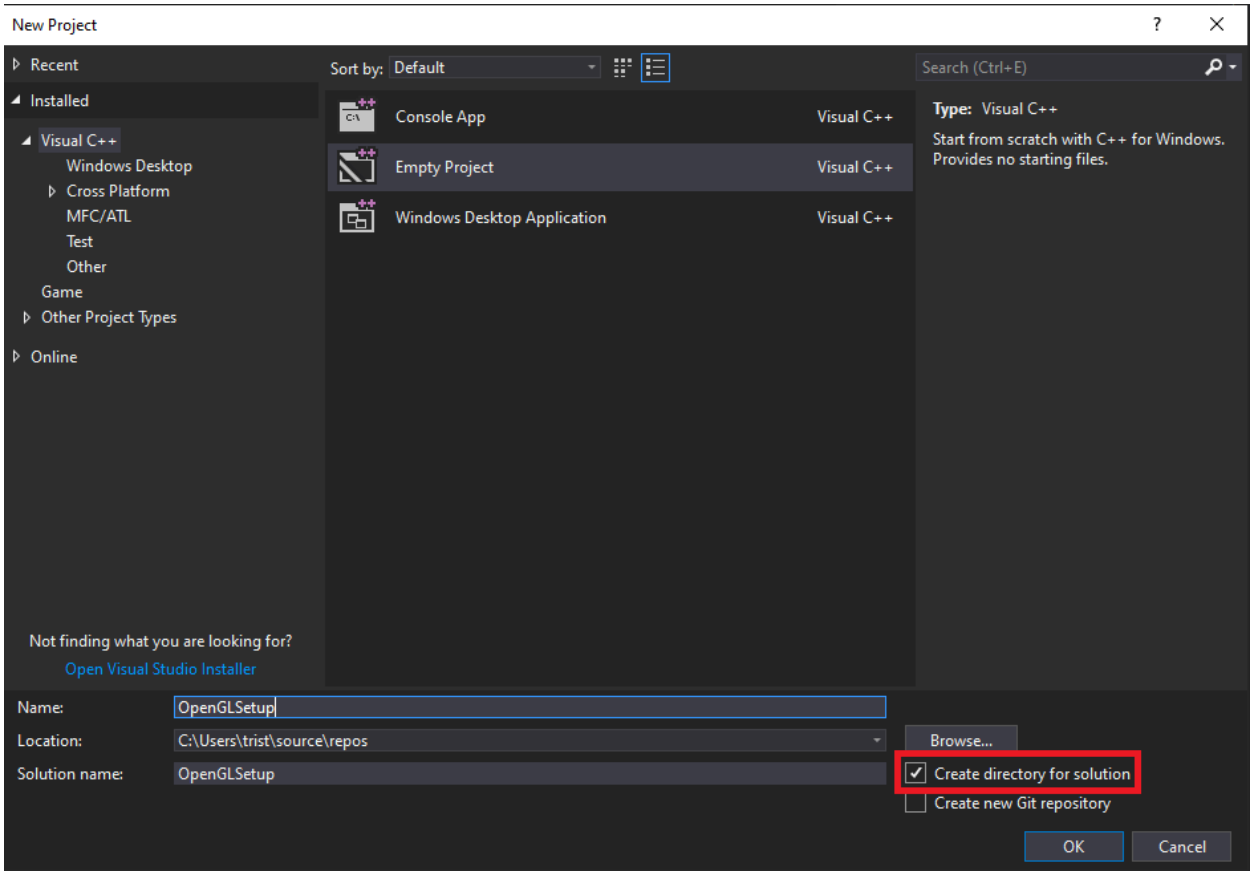
1) Create a new Project:



2) Select ***Empty Project***

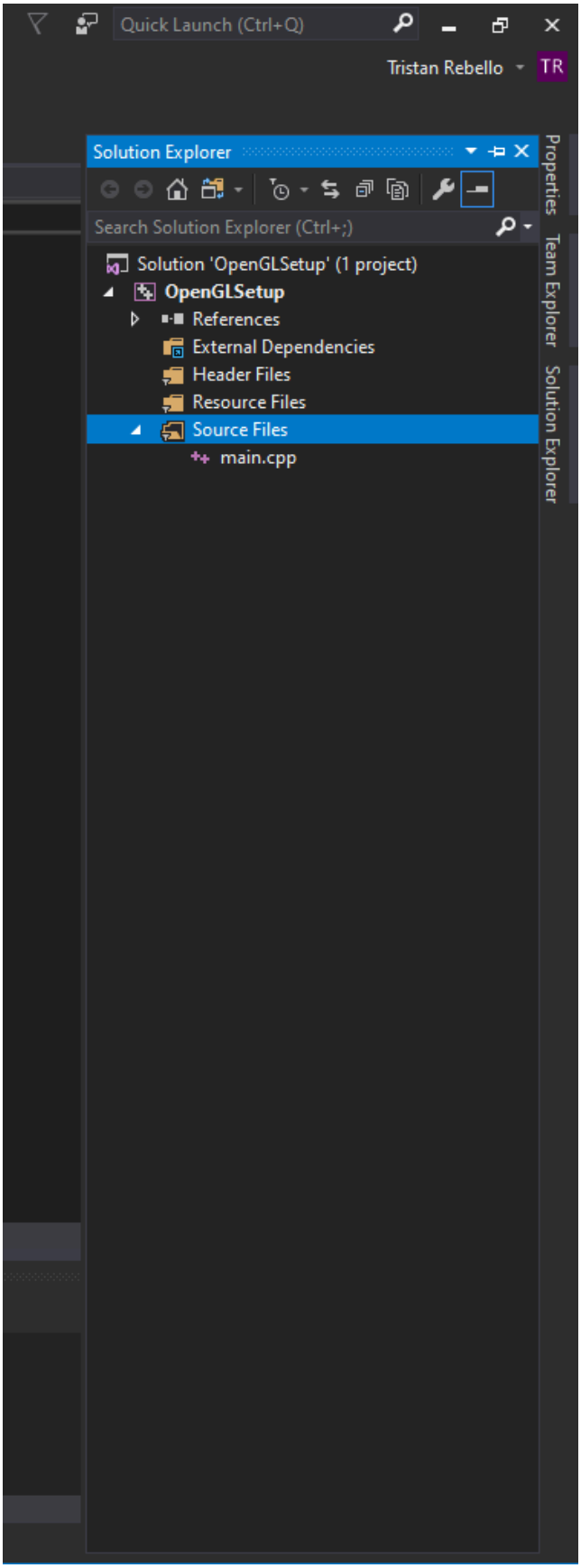


3) Make sure the 'Create directory for solution' box is checked

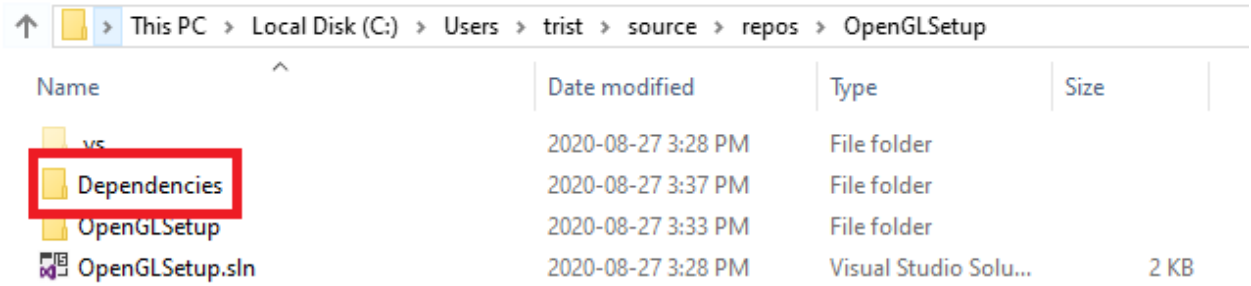


4) Remember where your project's location is. This will be called the Solution Directory. In this case it's ***C:\Users\trist\source\repos\OpenGLSetup***, which is my Solution directory. Then click OK.

5) Open up the Solution Explorer. It should be pinned on the right had side of the screen. Under the ***Source Files*** folder create a new C++ file called ***main.cpp***

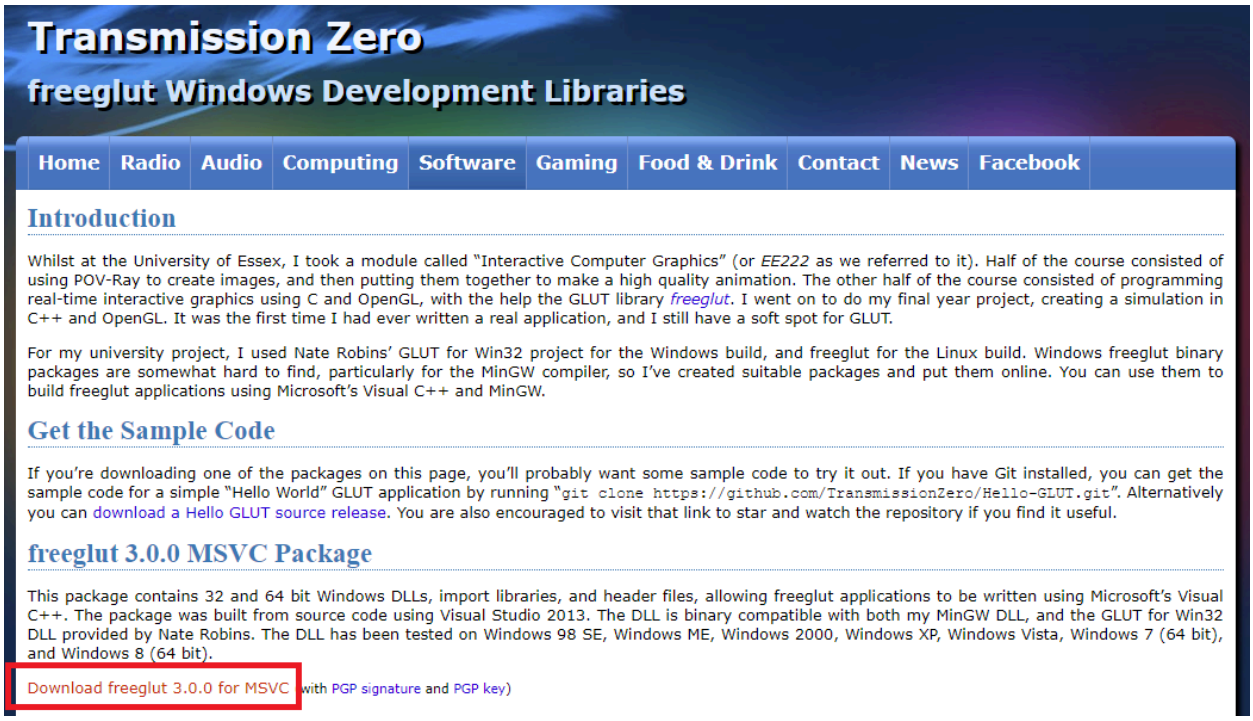


6) Navigate over to your Solution Directory and create a new folder called ***Dependencies***



FreeGLUT

To setup FreeGLUT, go to https://www.transmissionzero.co.uk/software/freeglut-devel/ and download the ***FreeGLUT 3.0.0 MSVC Package.***



Transmission Zero

freeglut Windows Development Libraries

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Introduction

Whilst at the University of Essex, I took a module called "Interactive Computer Graphics" (or *EE222* as we referred to it). Half of the course consisted of using POV-Ray to create images, and then putting them together to make a high quality animation. The other half of the course consisted of programming real-time interactive graphics using C and OpenGL, with the help the GLUT library *freeglut*. I went on to do my final year project, creating a simulation in C++ and OpenGL. It was the first time I had ever written a real application, and I still have a soft spot for GLUT.

For my university project, I used Nate Robins' GLUT for Win32 project for the Windows build, and freeglut for the Linux build. Windows freeglut binary packages are somewhat hard to find, particularly for the MinGW compiler, so I've created suitable packages and put them online. You can use them to build freeglut applications using Microsoft's Visual C++ and MinGW.

Get the Sample Code

If you're downloading one of the packages on this page, you'll probably want some sample code to try it out. If you have Git installed, you can get the sample code for a simple "Hello World" GLUT application by running "`git clone https://github.com/TransmissionZero/Hello-GLUT.git`". Alternatively you can download a Hello GLUT source release. You are also encouraged to visit that link to star and watch the repository if you find it useful.

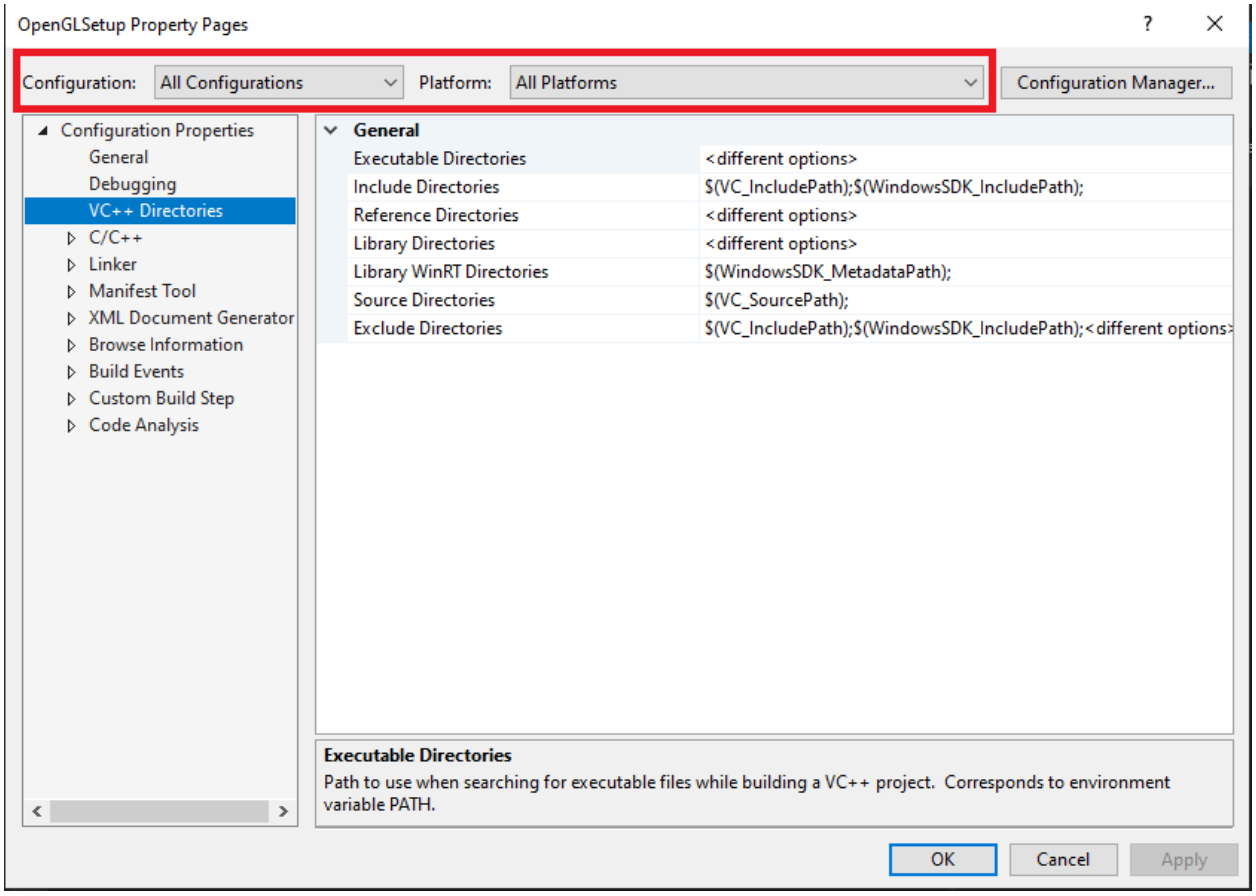
freeglut 3.0.0 MSVC Package

This package contains 32 and 64 bit Windows DLLs, import libraries, and header files, allowing freeglut applications to be written using Microsoft's Visual C++. The package was built from source code using Visual Studio 2013. The DLL is binary compatible with both my MinGW DLL, and the GLUT for Win32 DLL provided by Nate Robins. The DLL has been tested on Windows 98 SE, Windows ME, Windows 2000, Windows XP, Windows Vista, Windows 7 (64 bit), and Windows 8 (64 bit).

Download freeglut 3.0.0 for MSVC (with PGP signature and PGP key)

- Once downloaded, open the ZIP folder and copy the ***freeglut*** folder INTO your Dependencies directory.
- Go to ***Dependencies -> freeglut -> lib*** and create a new folder called ***Win32***. Move the file ***freeglut.lib*** into that folder.
- Go to ***Dependencies -> bin*** and create a new folder called ***Win32***. Move the file ***freeglut.dll*** into that folder

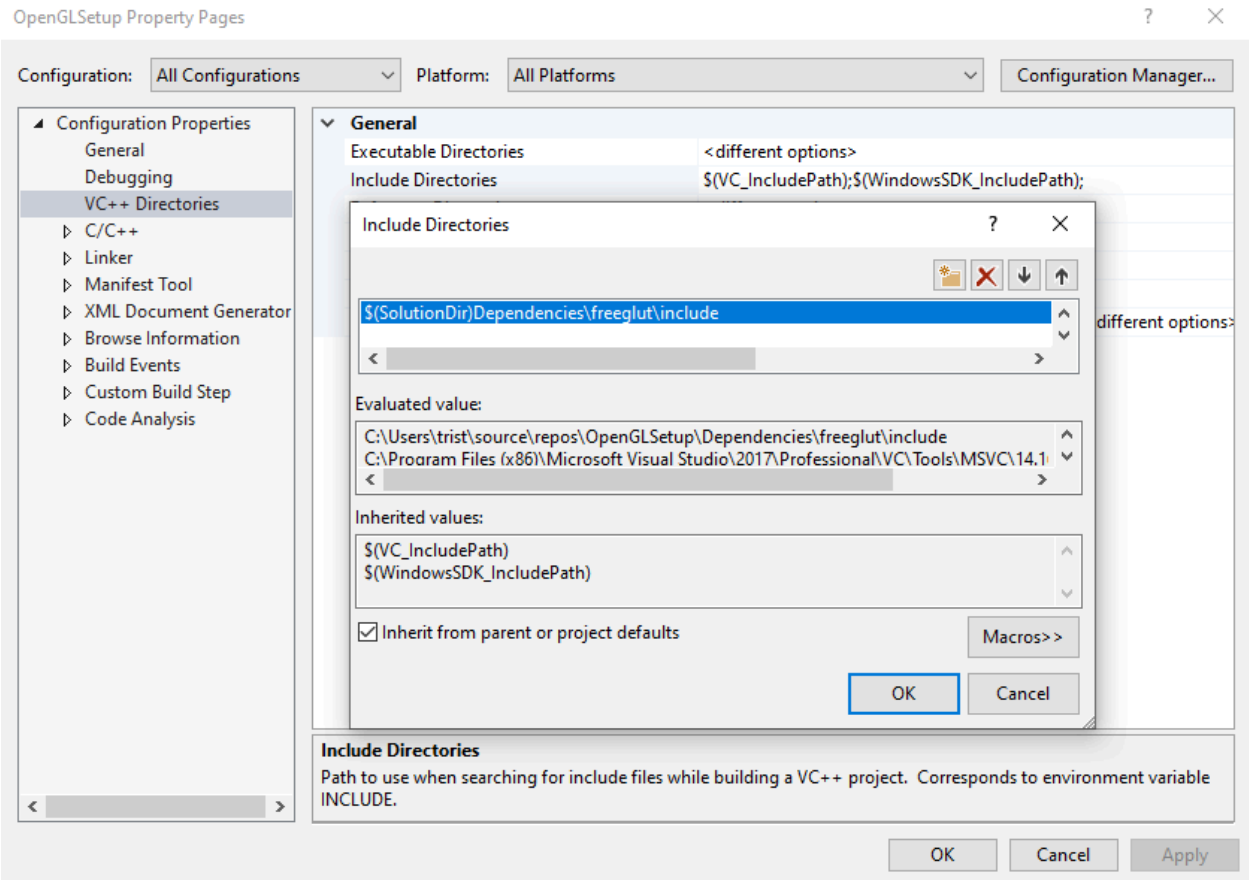
In the toolbar, head to ***Project -> Properties***. Make sure your Configuration and Platform settings are set to All Configurations and All Platforms respectively.



In ***VC++ Directories -> Include Directories*** type:

Correction (use absolute path):
C:\Users\b1ank\Desktop\OpenGLSetup\Dependencies\freeglut\include

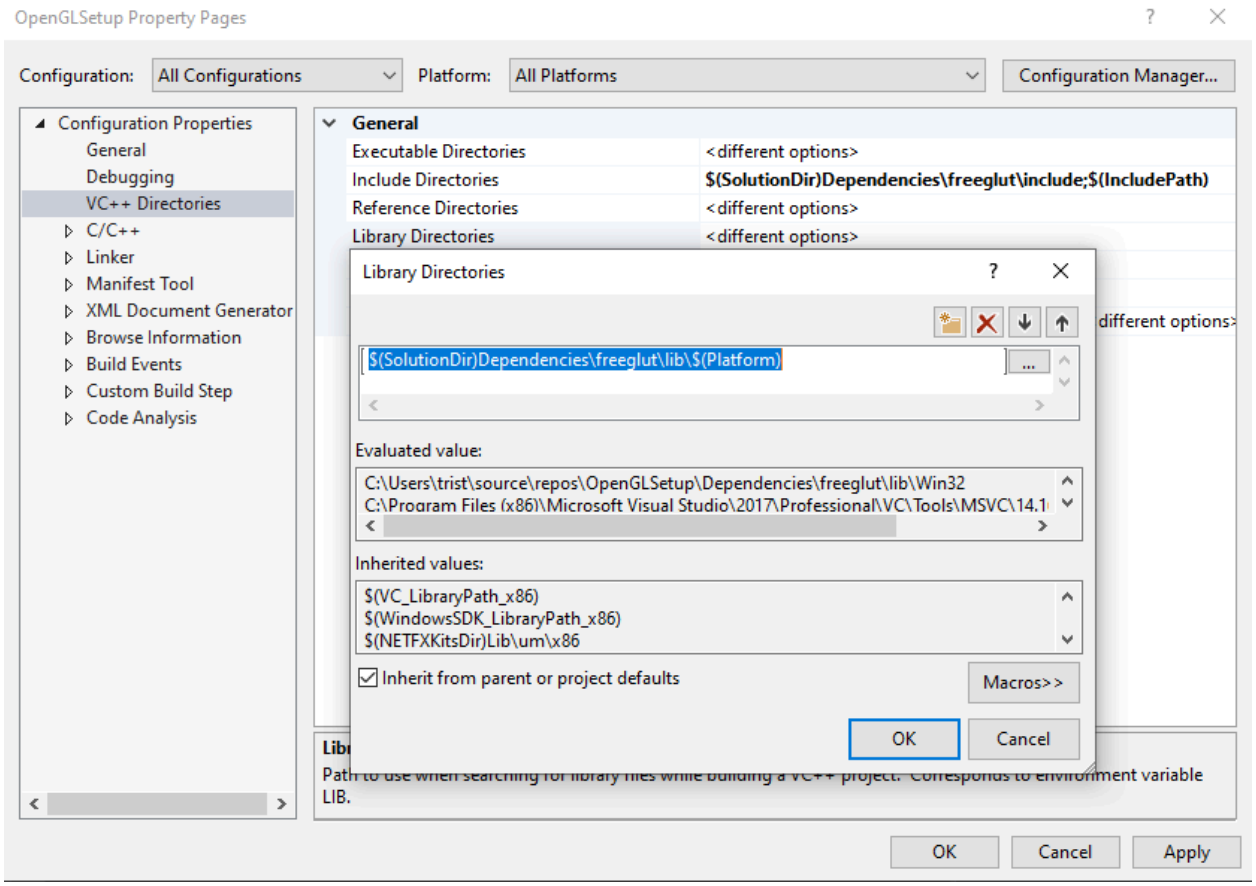
$(SolutionDir)Dependencies\freeglut\include. Hit OK.



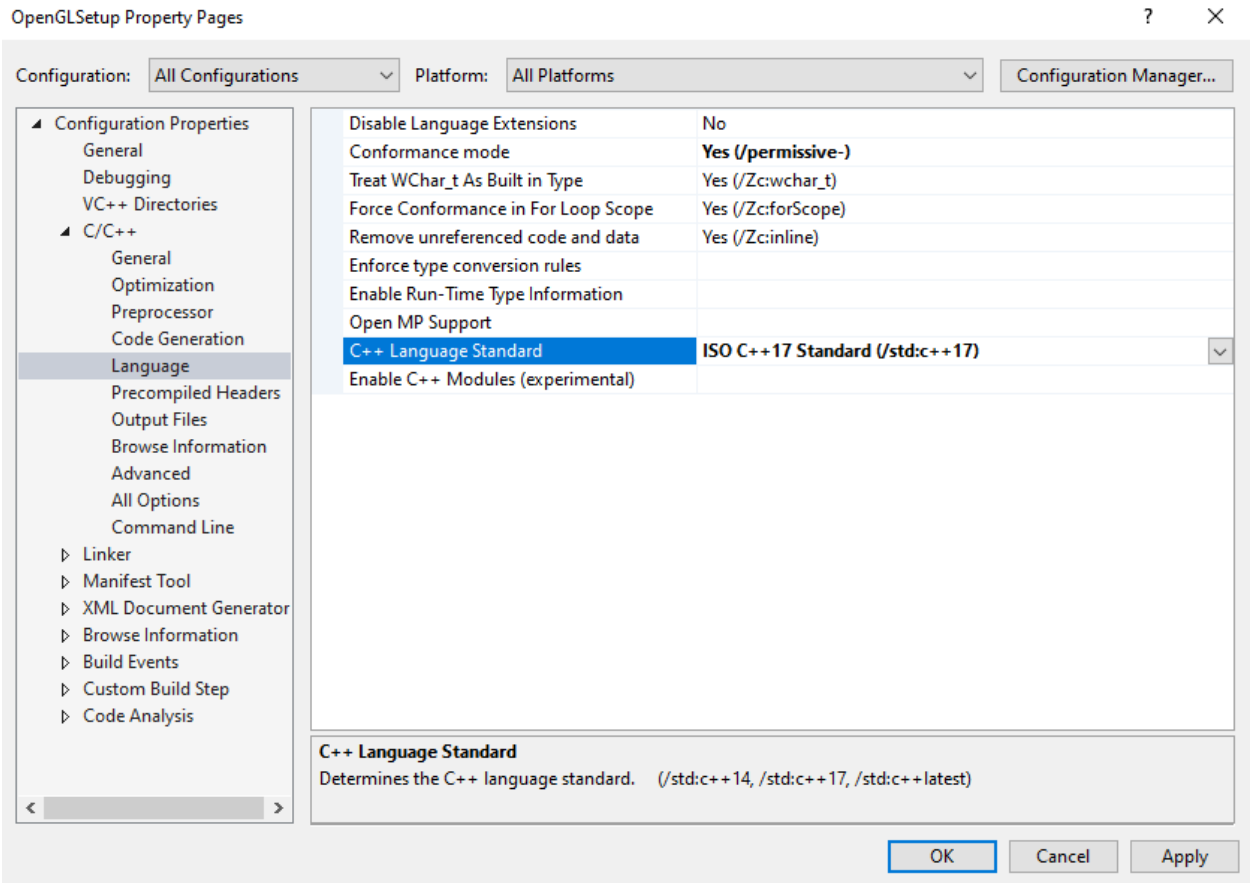
In ***VC++ Directories -> Library Directories***, type:

Correction (use absolute path):
C:\Users\b1ank\Desktop\OpenGLSetup\Dependencies\freeglut\lib\x64

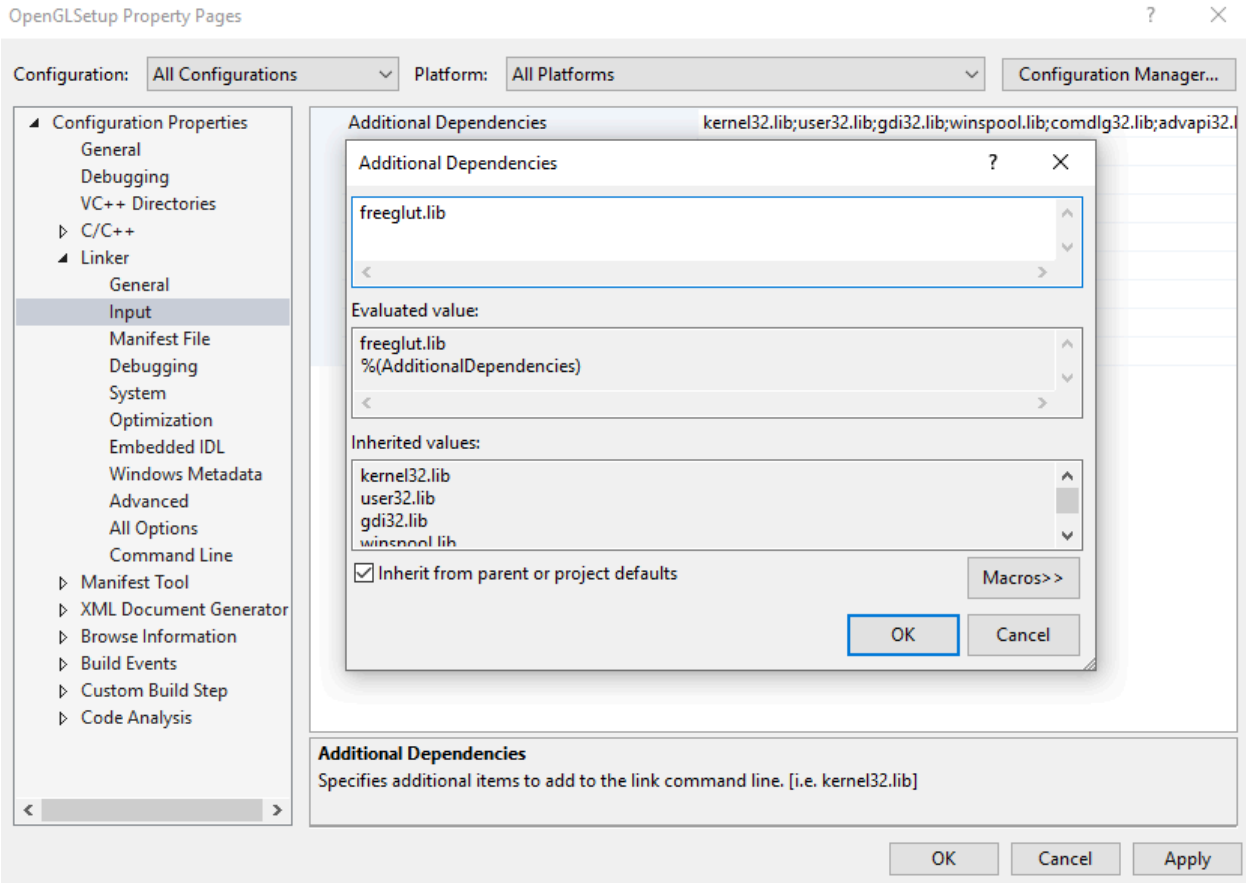
$(SolutionDir)Dependencies\freeglut\lib\$(platform). Hit OK



In ***C/C++ -> Language -> C++ Language Standard***, select the ***ISO C++17 Standard (/std:c++17)*** option



In ***Linker -> Input -> Additional Dependencies*** add ***freeglut.lib***. Hit OK.



Under ***Build Events -> Post-Build Event -> Command Line*** add

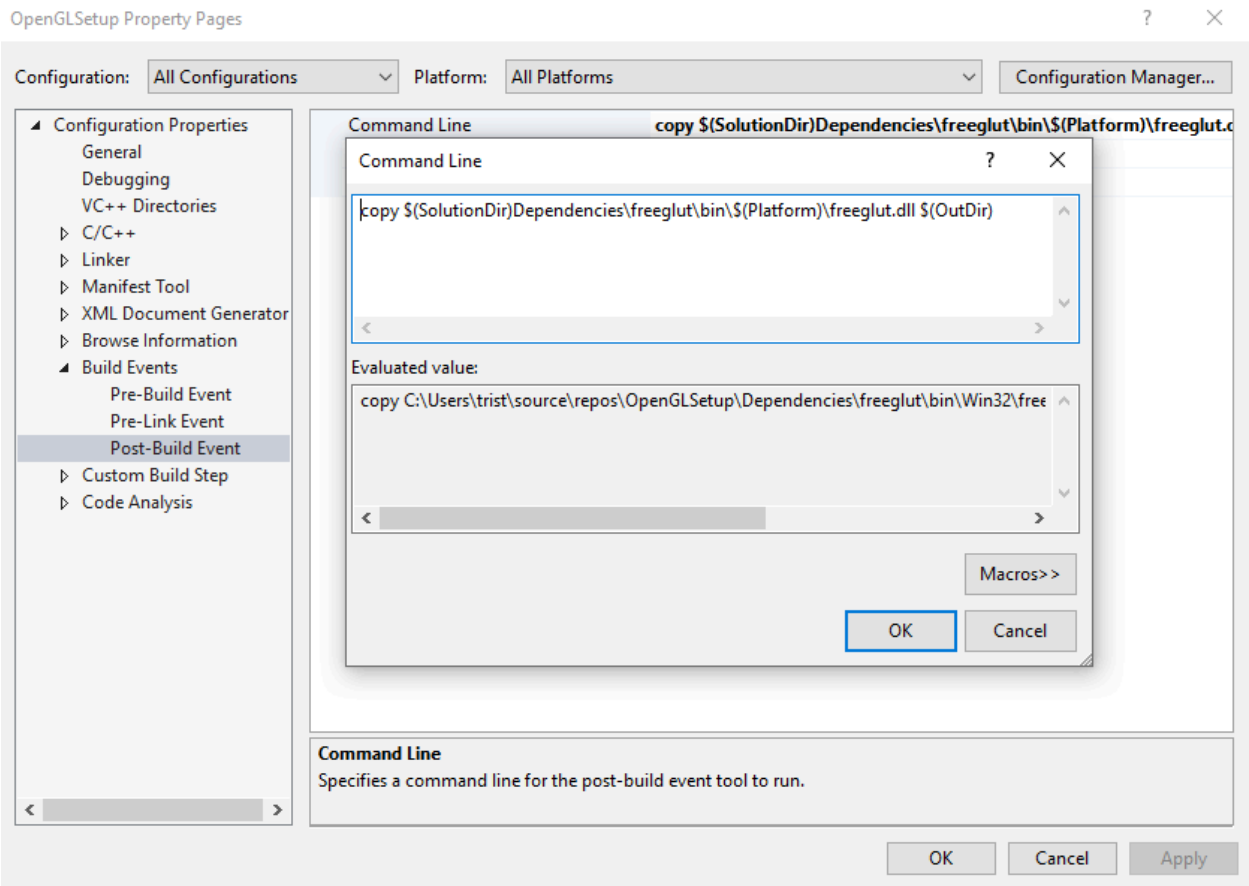
Correction:

copy
“C:\Users\b1ank\Desktop\OpenGLSetup\Dependencies\freeglut\lib\x64\freeglut.lib” “$(OutDir)”

copy
“$(SolutionDir)Dependencies\freeglut\bin\$(Platform)\freeglut.dll” “$(OutDir)”

Make sure there is one whitespace between freeglut.dll and $(OutDir)

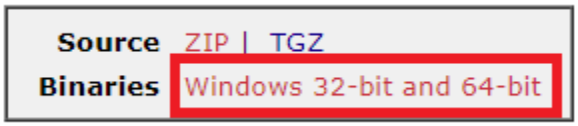
Hit OK and Click Apply.



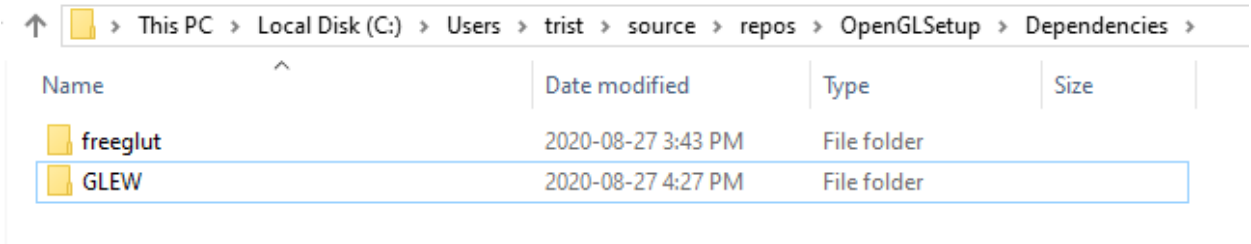
FreeGLUT should now be set up.

GLEW

To set up GLEW Go to http://glew.sourceforge.net/ and download the ***Windows 32-bit and 64-bit*** *binaries.*



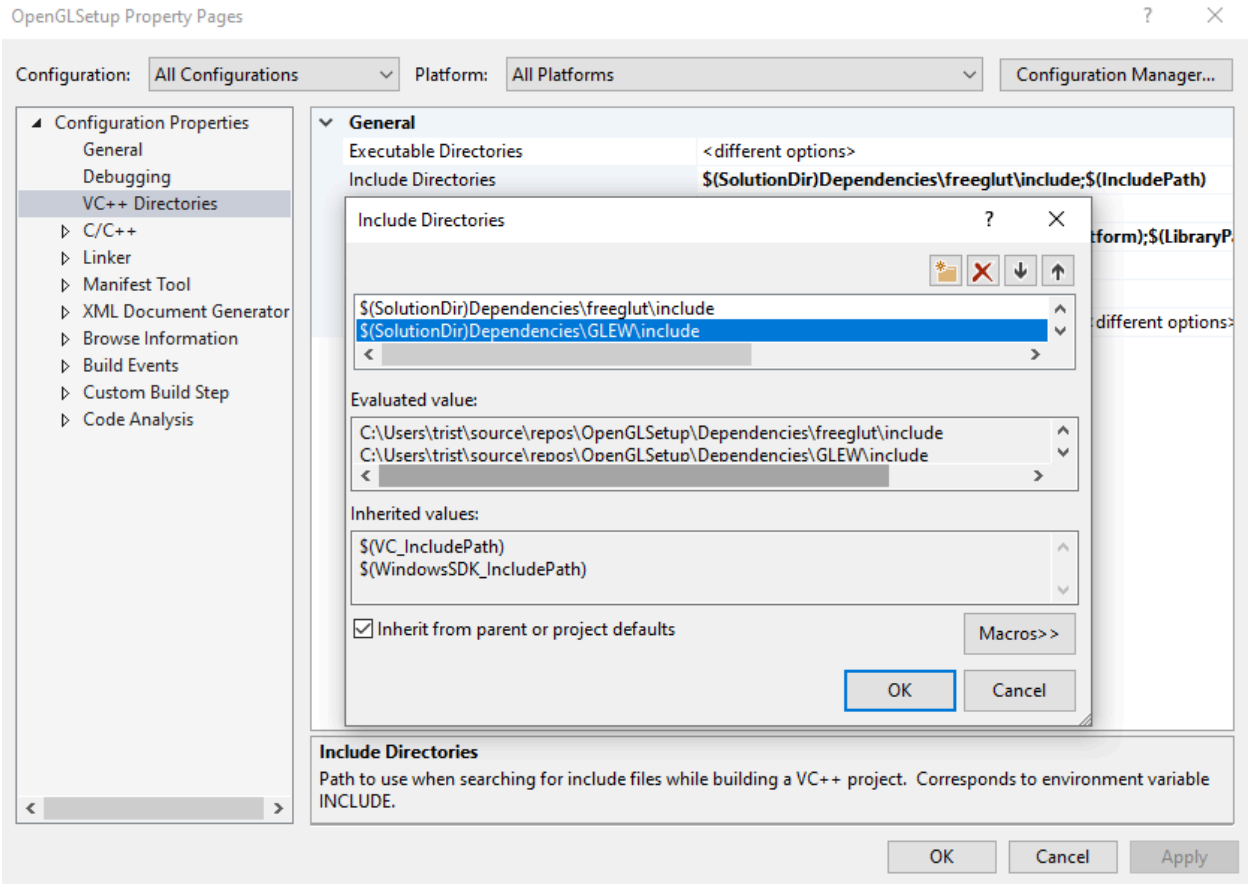
Open the downloaded ZIP and copy the ***glew-2.1.0*** into your Dependencies folder. Rename it to ***GLEW***



In the toolbar, head to ***Project -> Properties***. In ***VC++ Directories -> Include Directories*** type:

Correction:
C:\Users\b1ank\Desktop\OpenGLSetup\Dependencies\GLEW\include

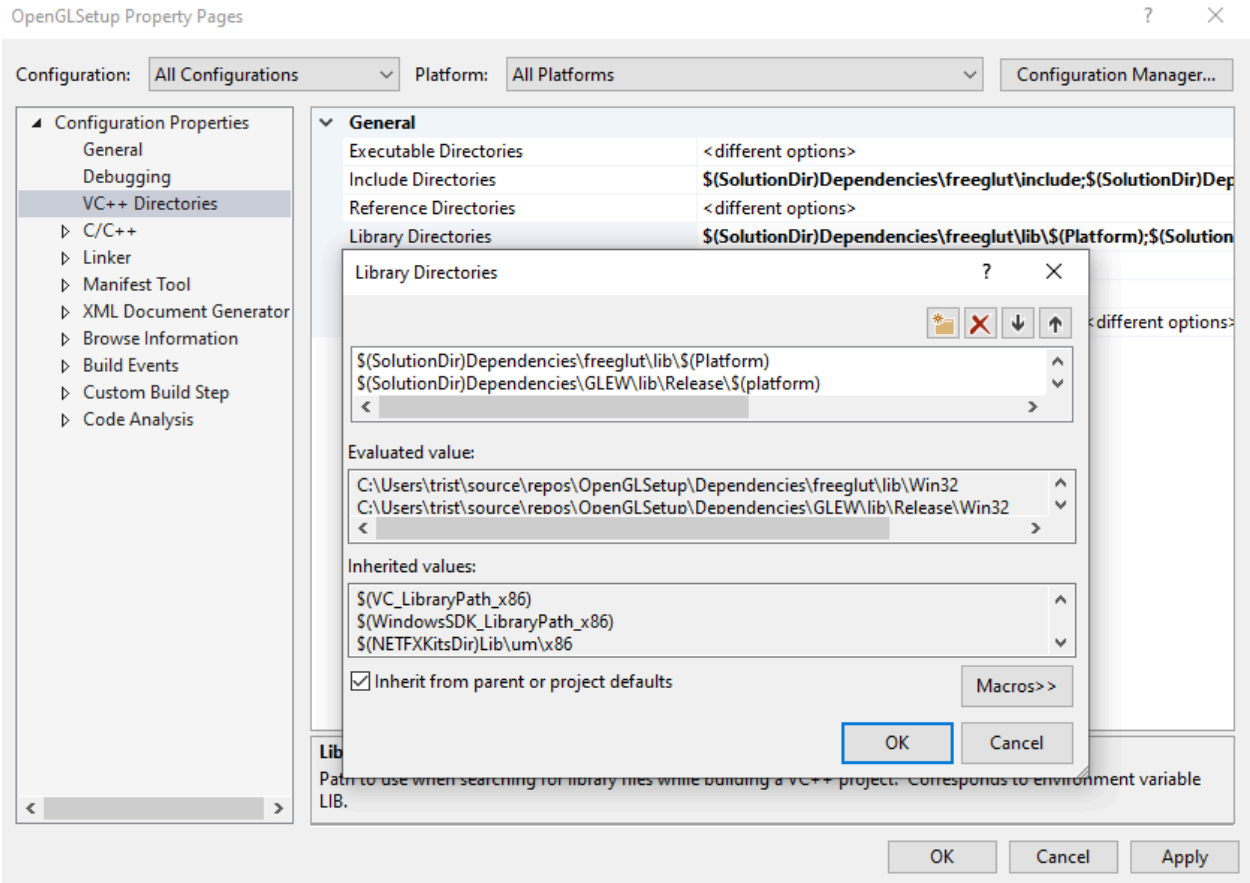
$(SolutionDir)Dependencies\GLEW\include. Hit OK.



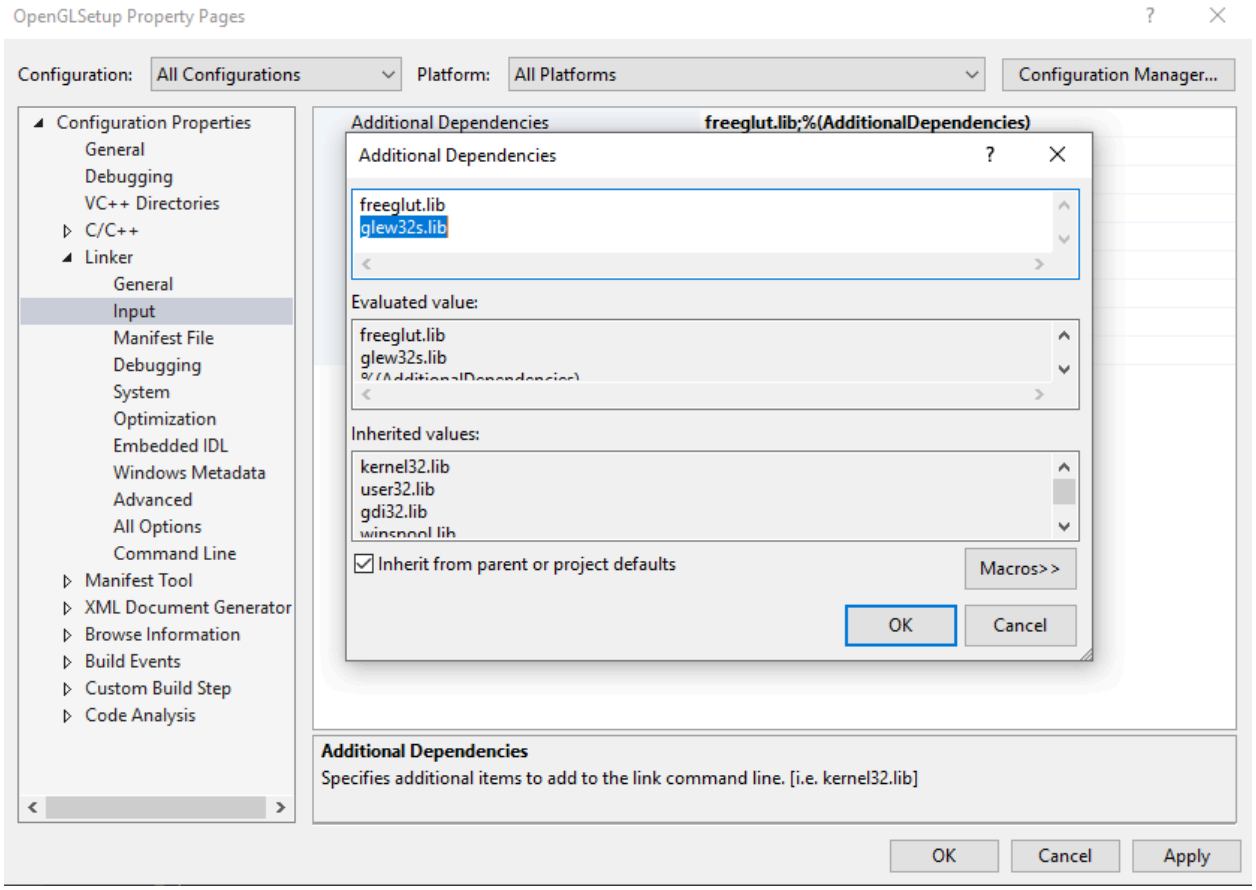
In ***VC++ Directories -> Library Directories***, type:

Correction:
C:\Users\b1ank\Desktop\OpenGLSetup\Dependencies\GLEW\lib\Release\x64

$(SolutionDir)Dependencies\GLEW\lib\Release\$(platform). Hit OK

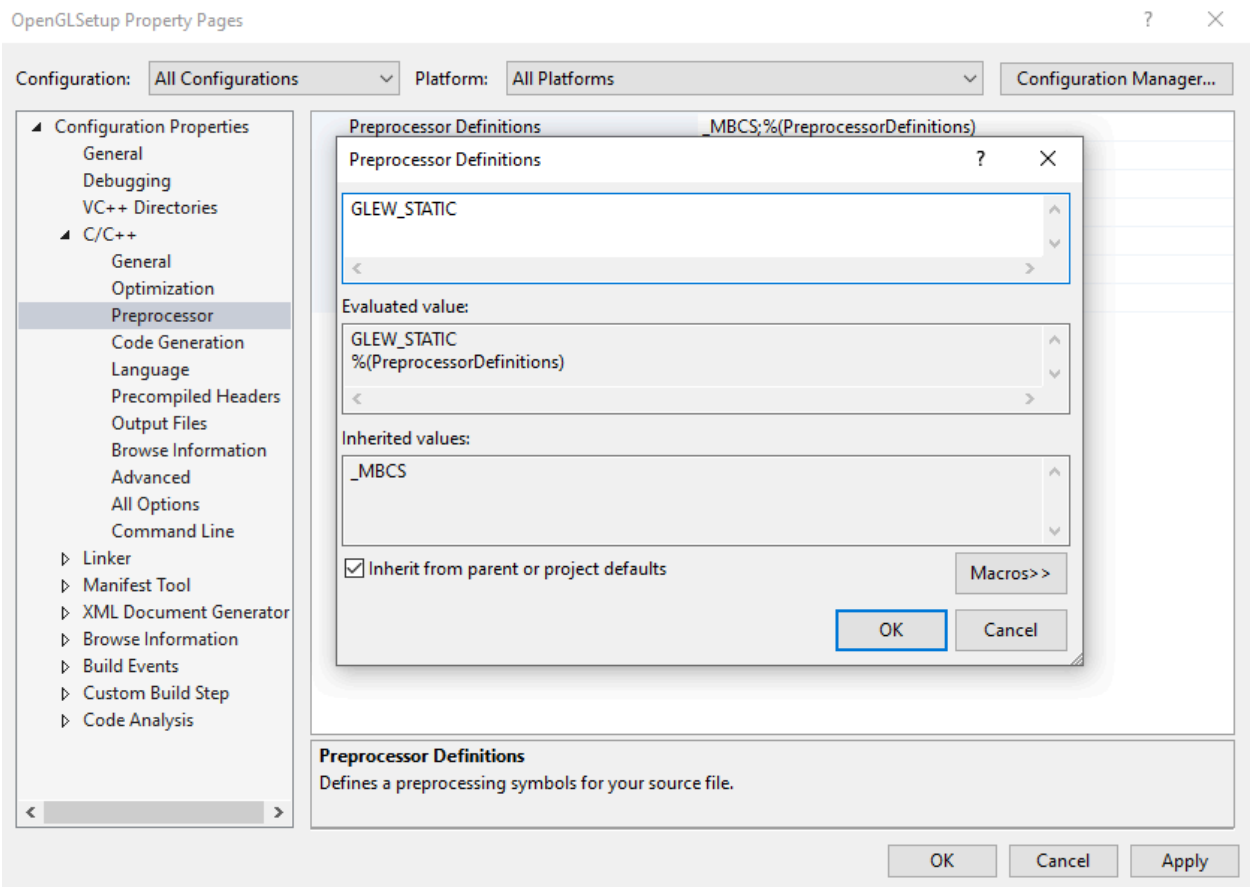


In ***Linker -> Input -> Additional Dependencies*** add ***glew32s.lib***. Hit OK



Lastly, In ***C/C++ -> Preprocessor -> Preprocessor Definitions*** add the ***GLEW_STATIC***. Hit OK. Hit Apply.

Also add inherit from parent or project defaults



Copy freeglut.dll from freeglut bin folder to the project folder (eg. OpenGlSetup)